Baby Safety Monitoring System

MPCA Project

TITLE OF THE PROJECT:

Team Members:

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Components used:

Hardware Components:

• Arduino UNO R3 – Microcontroller board used as the main processing unit
• DHT11 Temperature & Humidity Sensor – Measures ambient temperature around the baby
• 16x2 LCD Display (with I2C Module) – Displays real-time status messages (Baby Safe / Baby Moving / Temp readings)
• **P**IR Sensor (Infrared Proximity Sensor) – Detects baby movement/motion inside the crib
• Buzzer (Active) – Triggers an audio alert when high temperature or abnormal conditions are detected
• Breadboard (Full-size) – Used for prototyping and connecting all components without soldering
• Jumper Wires (Male-to-Male & Male-to-Female) – For electrical connections between components
• USB Cable (Type-B) – Powers the Arduino UNO from a laptop or power adapter

Software & Libraries:

• Arduino IDE – Development environment for writing and uploading code
• DHT Library – For reading temperature from the DHT11 sensor
• LiquidCrystal_I2C Library – For controlling the 16x2 LCD display via I2C protocol
• Wire Library – For I2C communication between Arduino and LCD module

Explanation about the project :

The Smart Room Monitoring System is an Arduino-based project designed to continuously monitor environmental conditions such as temperature, humidity, gas/smoke presence, and motion inside a room.

Working Principle:

The system uses a DHT11 sensor to measure temperature and humidity, a PIR sensor to detect human movement, and a gas sensor connected to an analog pin to identify smoke or harmful gases. At startup, the Arduino calibrates the gas sensor by calculating a baseline value of normal air conditions, which helps in accurately detecting abnormal gas levels later. During operation, the Arduino repeatedly reads data from all sensors, compares the values with predefined thresholds (for example, temperature above 34°C or gas level significantly higher than the baseline), and determines whether the environment is safe or unsafe. The 16x2 LCD with I2C module displays real-time information such as temperature, humidity, gas status (Smoke Alert or No Smoke), and motion status (M:Y or M:N). If any abnormal condition is detected—such as high temperature, gas leakage, or motion—the system activates a buzzer to alert the user immediately; otherwise, the buzzer remains off.

Key Features:

• Real-time temperature monitoring
• Motion detection
• 16x2 LCD display for instant visual status updates
• Buzzer alert for high-temperature emergencies
• This system is efficient, low-cost, and reliable
•making it suitable for applications like home automation, fire and safety systems

Circuit Diagram:

Pseudocode:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <DHT.h>
```

```
LiquidCrystal_I2C lcd(0x27, 16, 2);

// Pins
#define DHTPIN 2
#define DHTTYPE DHT11
#define PIR 3
#define GAS A0
#define BUZZER 7

DHT dht(DHTPIN, DHTTYPE);

// Gas calibration
int baseValue = 0;
bool gasAlert = false;

void setup() {
  lcd.init();
  lcd.backlight();

  pinMode(PIR, INPUT);
  pinMode(BUZZER, OUTPUT);

  digitalWrite(BUZZER, HIGH); // OFF

  dht.begin();

  lcd.print("Calibrating...");

  for (int i = 0; i < 50; i++) {
    baseValue += analogRead(GAS);
    delay(100);
  }
  baseValue = baseValue / 50;

  lcd.clear();
  lcd.print("System Ready");
  delay(2000);
  lcd.clear();
}

void loop() {
```

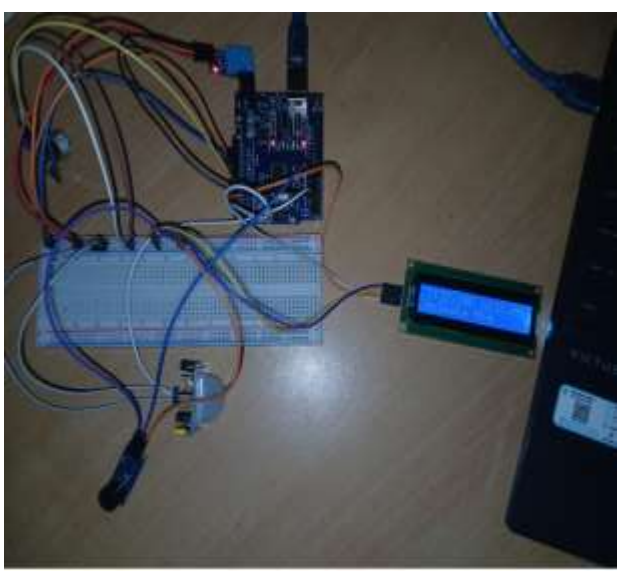
```
int motion = digitalRead(PIR);
int gasValue = analogRead(GAS);
float temp = dht.readTemperature();
float hum = dht.readHumidity();

bool tempAlert = false;

lcd.clear();

// -------- LINE 1 (TEMP + HUM) --------
lcd.setCursor(0, 0);

if (isnan(temp) || isnan(hum)) {
  lcd.print("Sensor Error");
} else {
  lcd.print("T:");
  lcd.print(temp);
  lcd.print("C H:");
  lcd.print(hum);
  lcd.print("%");

  if (temp > 34) {
    tempAlert = true;
  }
}

// -------- GAS --------
if (gasValue > baseValue + 70) {
  gasAlert = true;
}
else if (gasValue < baseValue + 30) {
  gasAlert = false;
}

// -------- LINE 2 --------
lcd.setCursor(0, 1);

if (gasAlert) {
  lcd.print("Smoke ALERT ");
} else {
  lcd.print("NoSmoke ");
}
```

```
  if (motion == HIGH) {
    lcd.print("M:Y");
  } else {
    lcd.print("M:N");
  }

  // -------- BUZZER --------
  if (gasAlert || motion == HIGH || tempAlert) {
    digitalWrite(BUZZER, LOW);   // ON
  } else {
    digitalWrite(BUZZER, HIGH);  // OFF
  }

  delay(500);
}
```

Output Pictures with explanations:

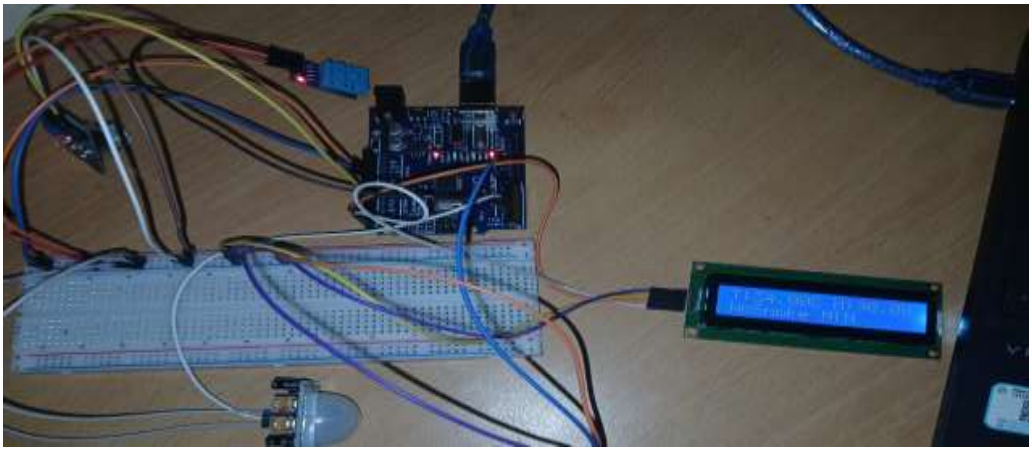
Output 1 -Normal Operating Condition

When all sensor readings are within safe limits—temperature is below the threshold, no gas/smoke is detected, and no motion is sensed—the system functions normally, displaying safe status on the LCD and keeping the buzzer off.



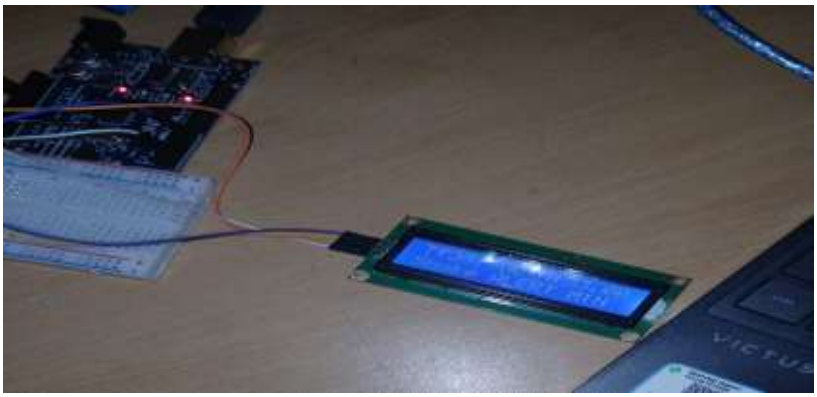
Output 2 – Buzzer For high temperature

The DHT11 sensor measures temperature using an internal thermistor, converts it into a digital signal, and sends the data to the Arduino for processing and display on the LCD.



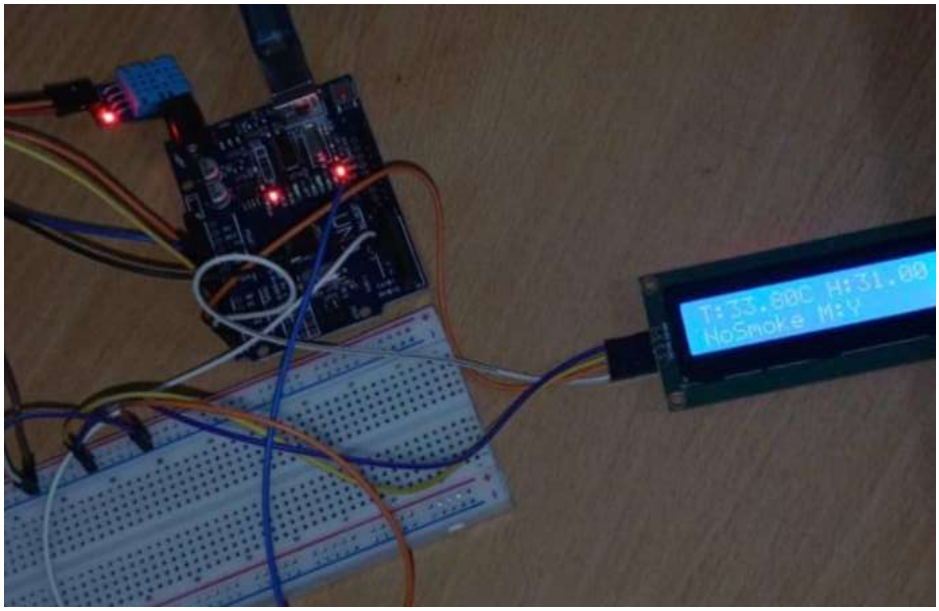
Output 3 – Buzzer if smoke is preset

When smoke or gas is detected above the threshold level, the system activates the buzzer to provide an immediate warning, alerting users to potential danger in the environment.



Output 4 – Buzzer if motion is there

When motion is detected by the PIR sensor, the system activates the buzzer to alert the user about movement or presence in the monitored area.


T:33.00C H:31.00
NoSmoke M:Y